SAVEETHA SCHOOL OF ENGINEERING
C PROGRAMMING ASSIGNMENT – Grama Jyothi Billing Program

Field	Entry
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Register Number	1912260562
Course Code & Name	CSA0209 C PROGRAMMING
Slot	B
Assignment / Task Title	Design, Implement and Validate a Reliable Billing Program
GitHub Repository URL	https://github.com/bavadarini28 /CSA0209-1912260562-BAVADARANI.S.git
Submission Date	12 NOVEMBER 2026

TASK 8 – Trace the Compilation Process

Aim

To follow billing.c through each compilation stage.

```
gcc -E billing.c -o billing.i    # preprocessed: 866 lines (header + stdio.h expanded)
gcc -S billing.c -o billing.s    # assembly: 279 lines
gcc -c billing.c -o billing.o    # object file: 3,080 bytes
gcc billing.c -o billing         # linked executable: 16,224 bytes
```

The preprocessor removed all comments and #include/#define lines, pasted in stdio.h and tariff.h and replaced every macro: in billing.i the line reads if (u > 200) { paise += (long)(u - 200) * 450; … }, with no trace of SLAB2_UNITS or RATE3_PAISE.

Error type	Example from this work	When detected
Compile-time	Missing semicolon or undeclared name, e.g. using RATE4_PAISE	gcc -c, before an object file exists
Link-time	Calling compute_bil() (typo) – undefined reference	Linking step (ld)
Run-time	Dividing by zero, or an infinite loop on bad scanf input before the flush was added	While the program runs

Warnings: with -Wall -Wextra, gcc reported "format %d expects argument of type int, but argument has type long int" for printf("%d", v) and unused-variable warnings. Resolved by using %ld and removing/using the variable. The final billing.c compiles with no warnings.

Alternative Concept – separate files (input.c, tariff.c, report.c)

Build: gcc -c input.c tariff.c report.c, then gcc *.o -o billing (or a Makefile). Compile time after a change to one file: single file – the whole program, O(N) in total code size; separate – only the changed file is recompiled plus a fast link, O(n_i + link). Space: the object files together are slightly larger than one (extra symbol tables and duplicated header declarations), e.g. three .o files may total a few KB more than one, and all are kept on disk. For a program this small, the difference is negligible, but separation scales better.

Result: Each stage observed; errors and warnings classified.